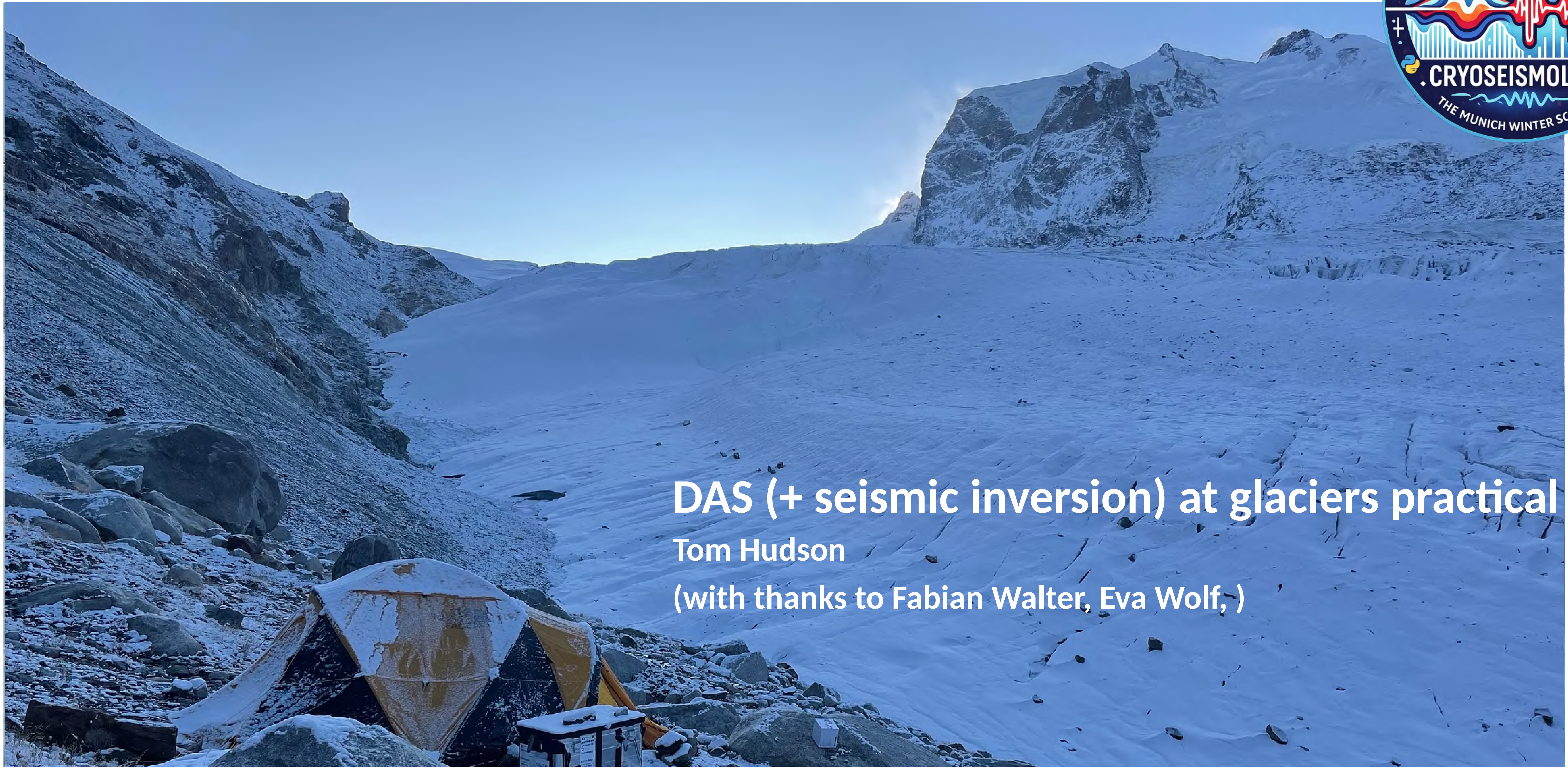
ETH zürich



DAS (+ seismic inversion) at glaciers practical

Tom Hudson
(with thanks to Fabian Walter, Eva Wolf,)

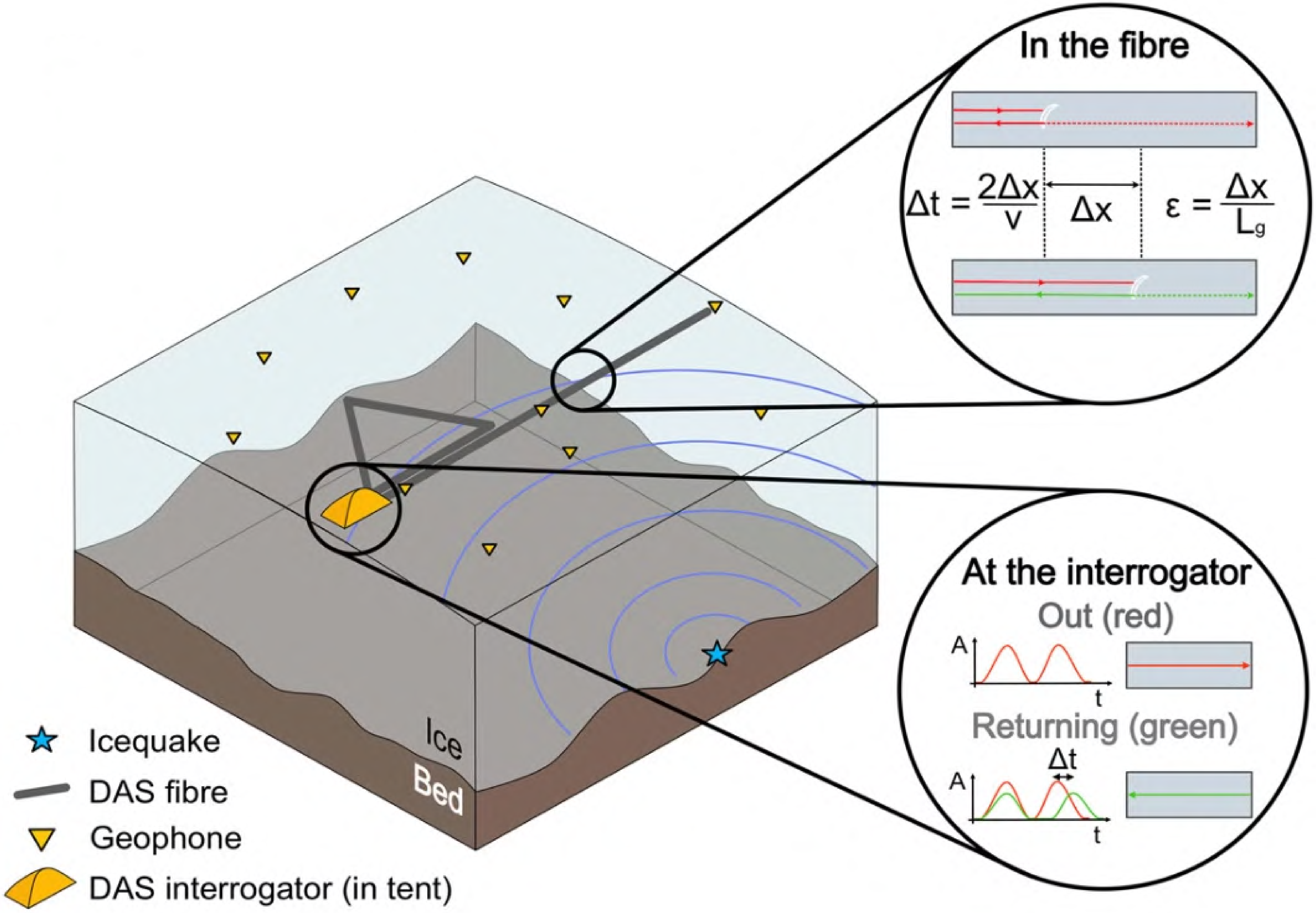
Overview of practical

1. Some context

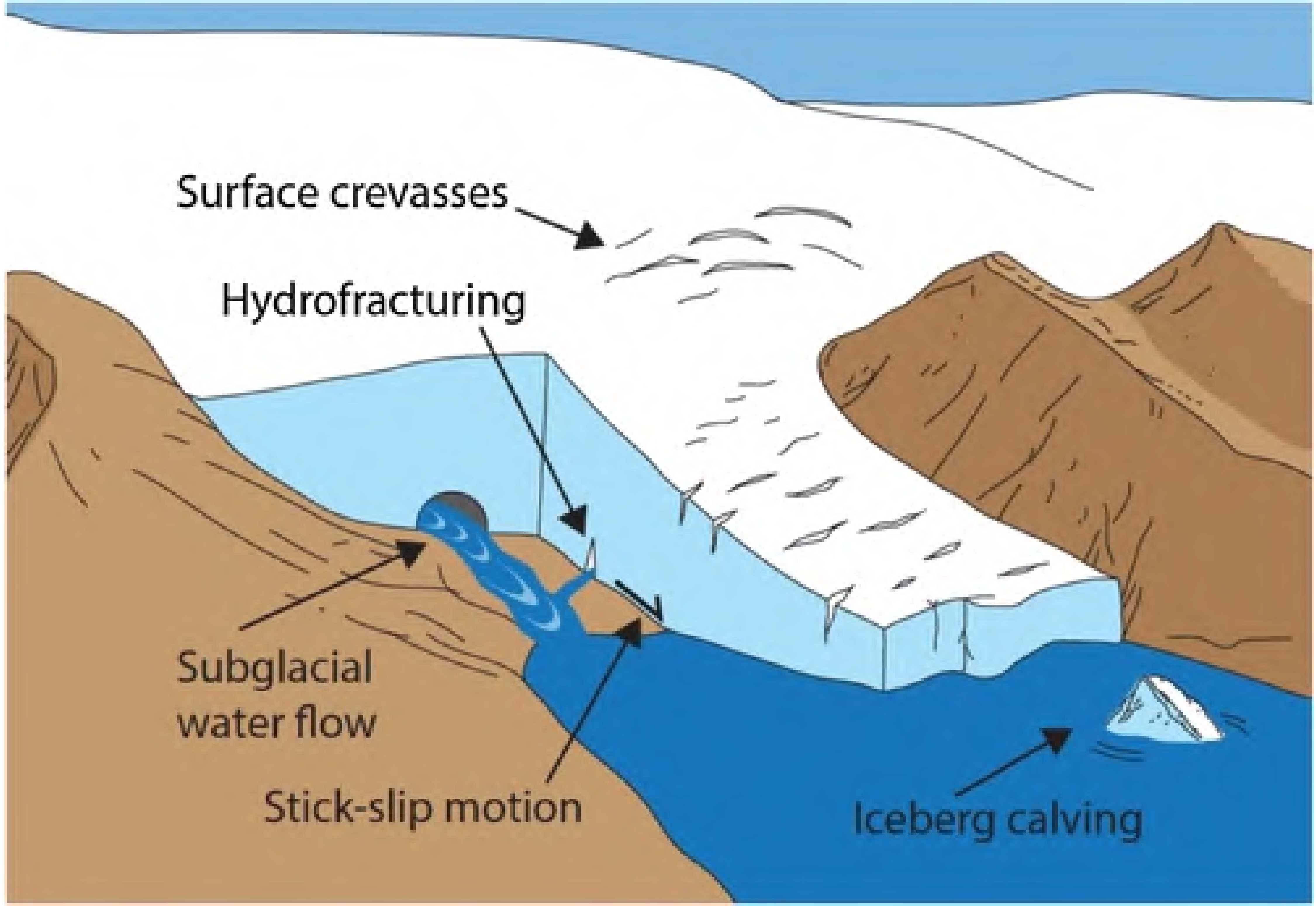
2. The dataset

3. Choice of two different inversion practicals
    Source inversion
    “Tomographic” inversion

Distributed Acoustic Sensing (DAS)

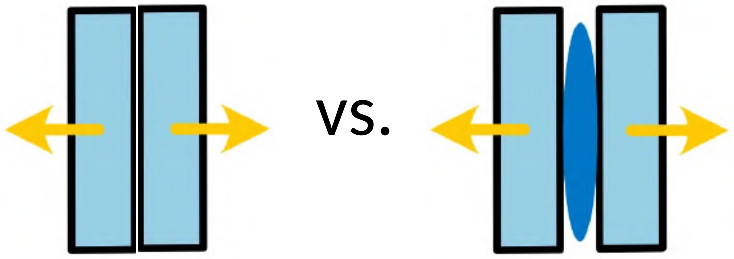


Two glacier processes that we are interested in



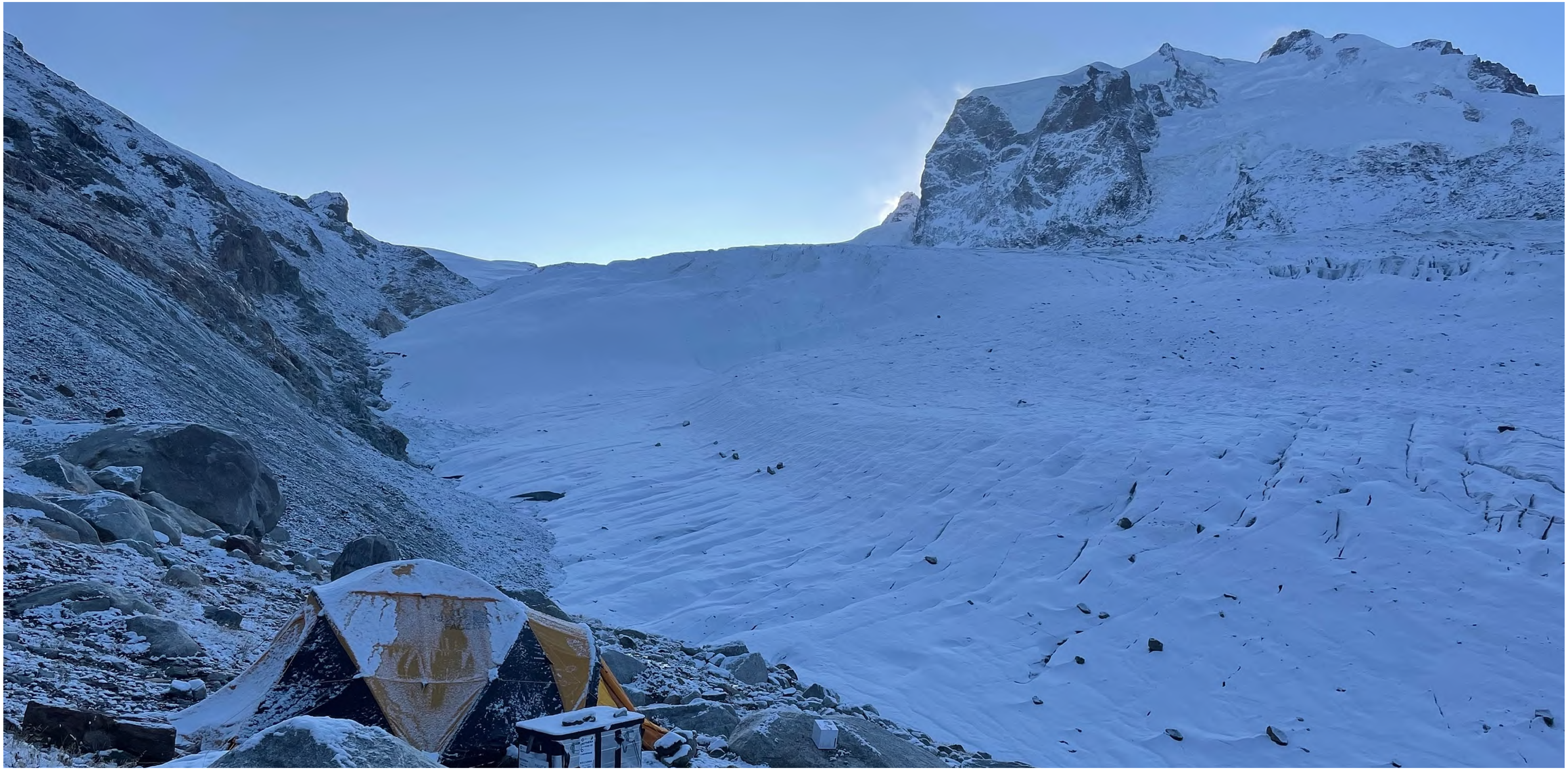
Podolskiy and Walter (2016)

1. Crevasse fracture mechanism



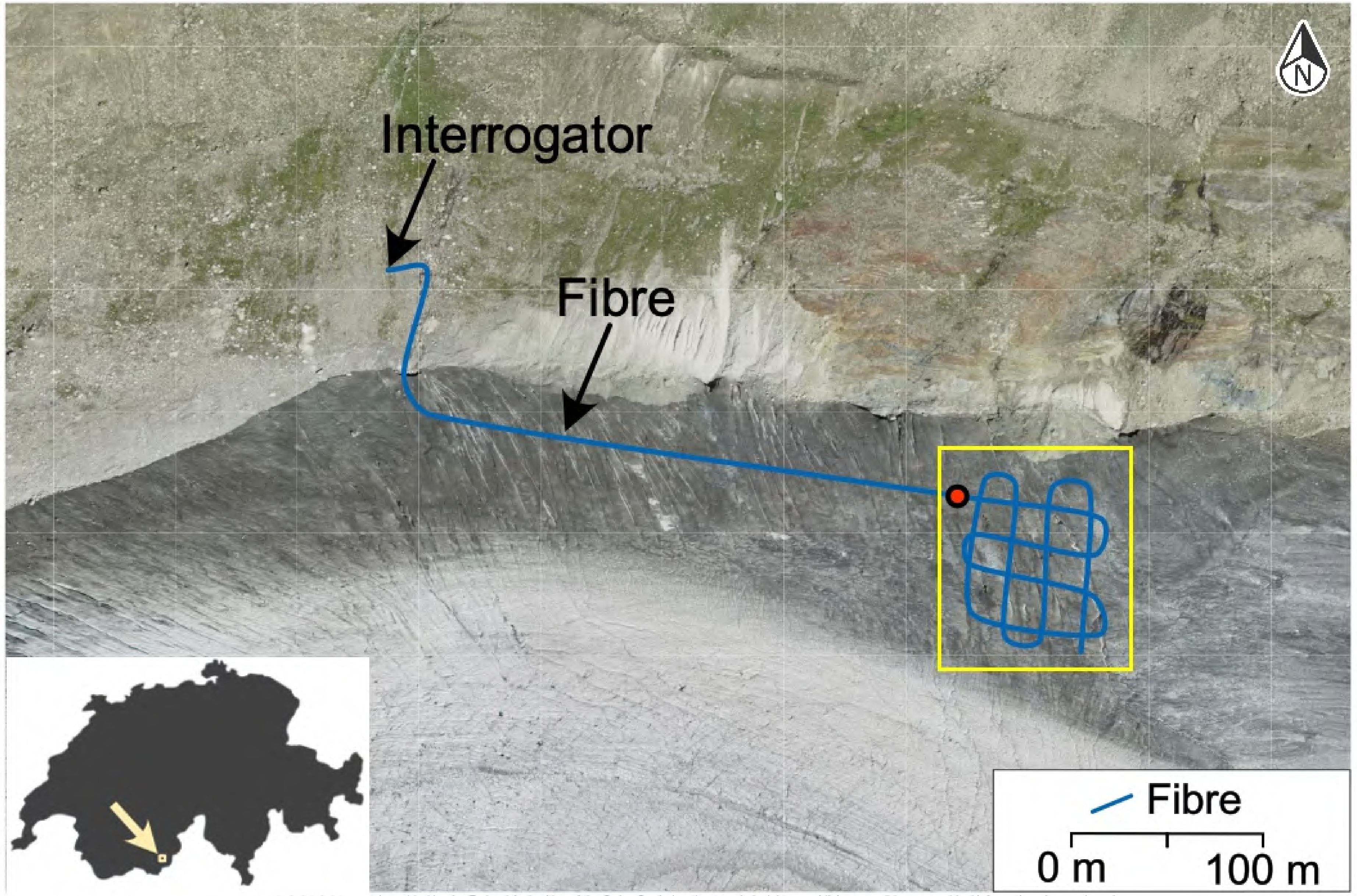
2. Subsurface fracture extent

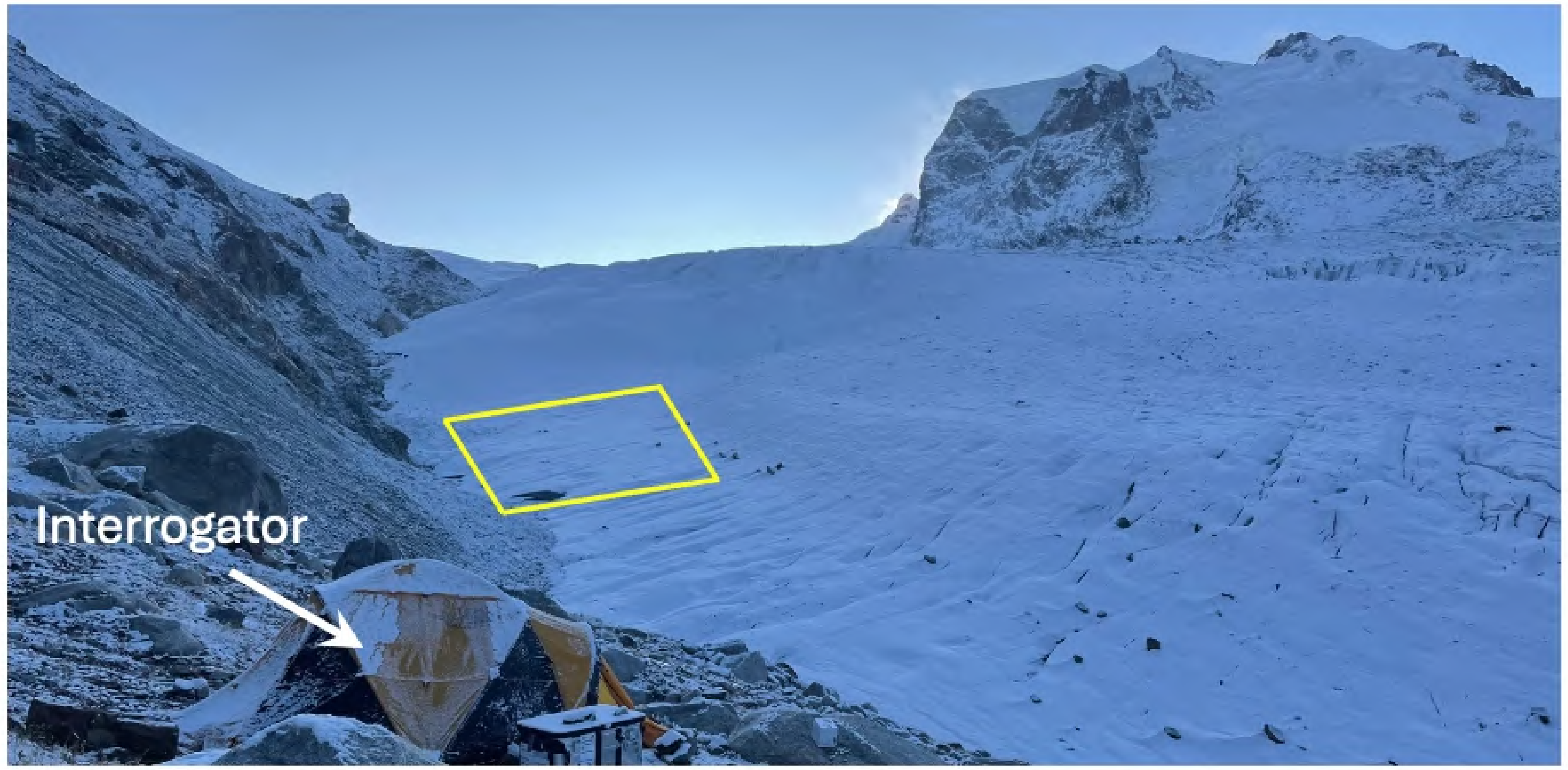
The dataset



The dataset

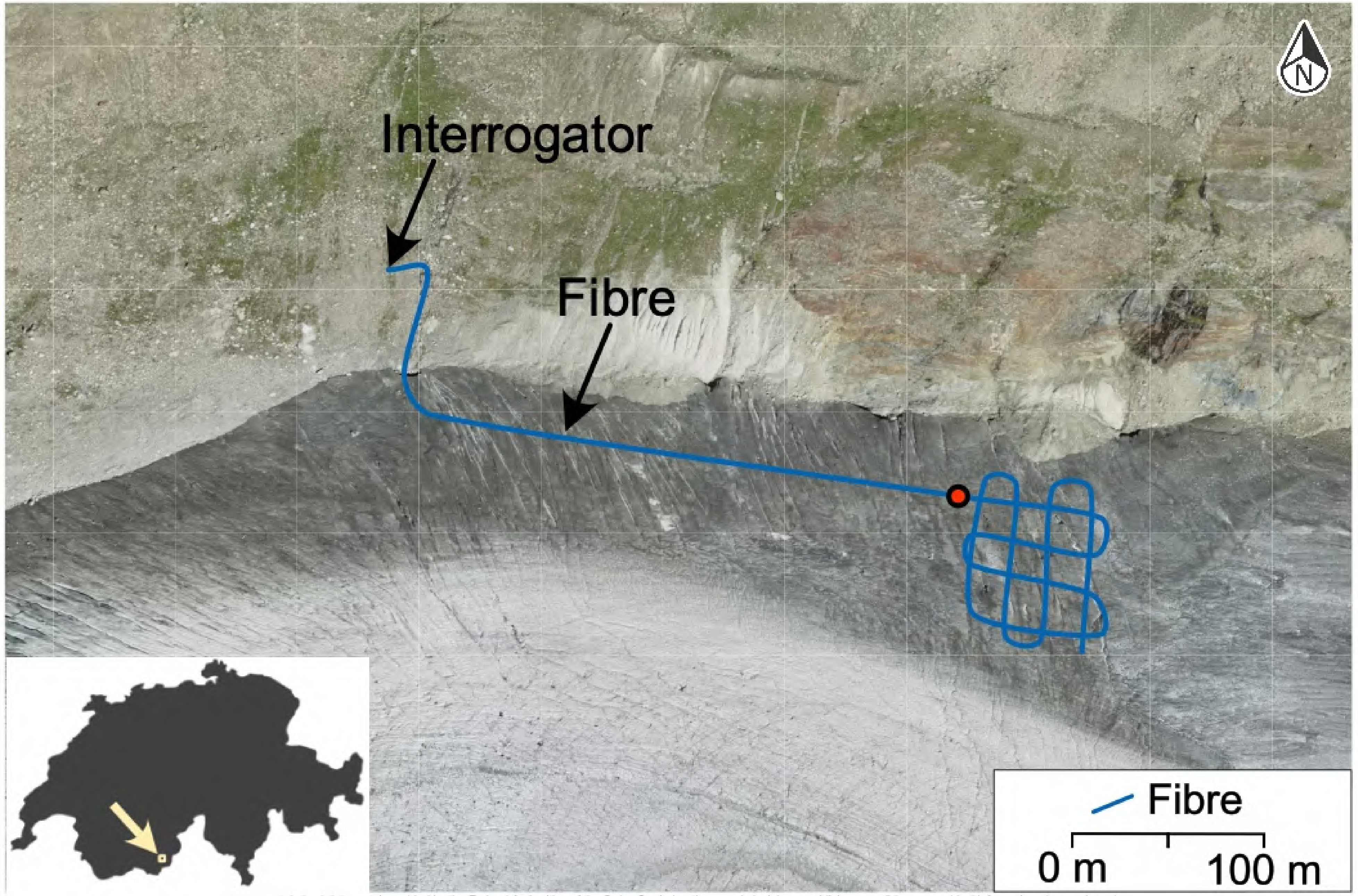
Location: Gornergletscher, Swiss Alps
Fibre length: 1 km
Interrogator: Sintela Onyx (with 6 m gauge length, sampling @1000 Hz)

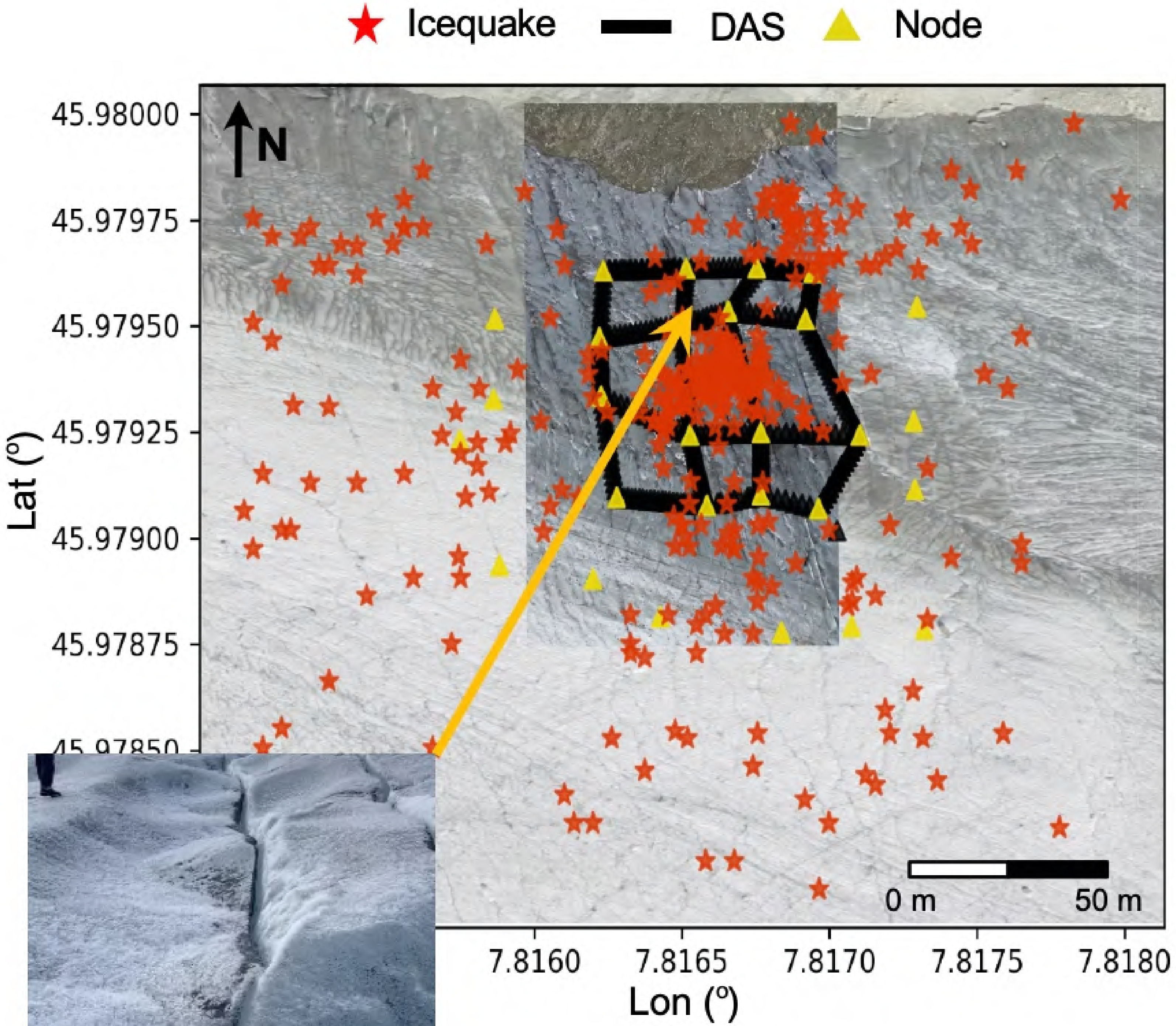




The experiment

Location: Gornergletscher, Swiss Alps
Fibre length: 1 km
Interrogator: Sintela Onyx (with 6 m gauge length, sampling @1000 Hz)





Water inside!

This practical:

Two options

Icequake source mechanism inversion

Surface-wave velocity anisotropy inversion

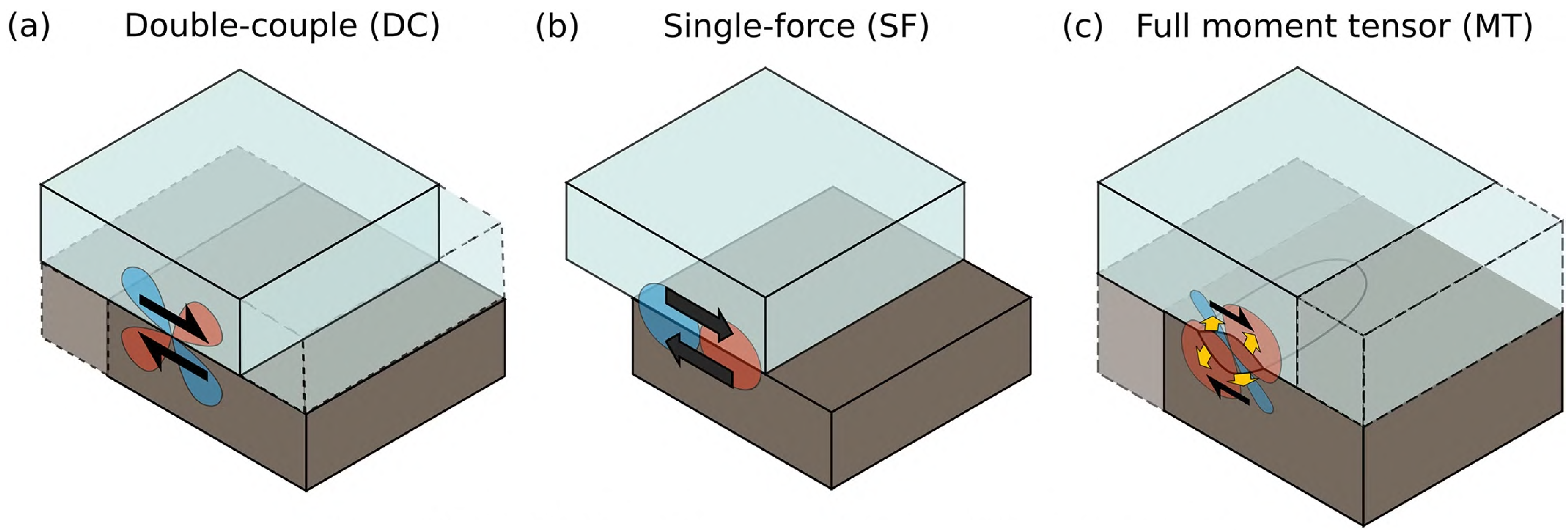
Seismic source mechanism inversion: Introduction

1. **Source mechanisms (moment tensors)**
2. Data
3. Inversion setup
4. Nuances of fibreoptics

Alvizuri et al. (2018)

Source mechanisms

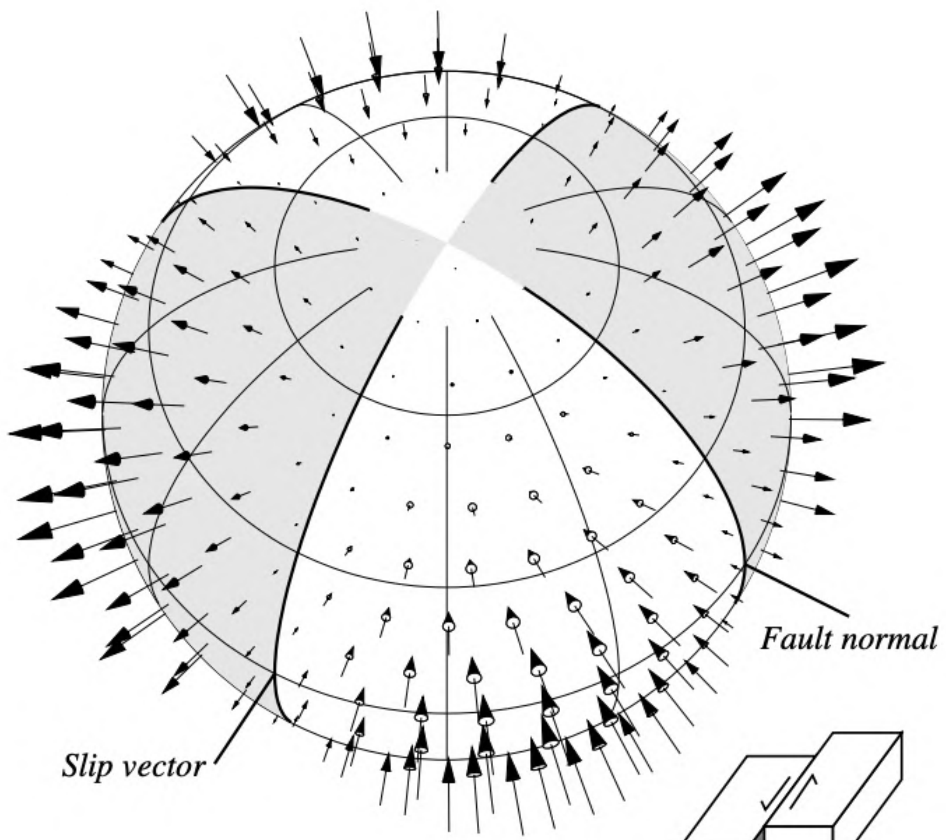
Fundamental types:



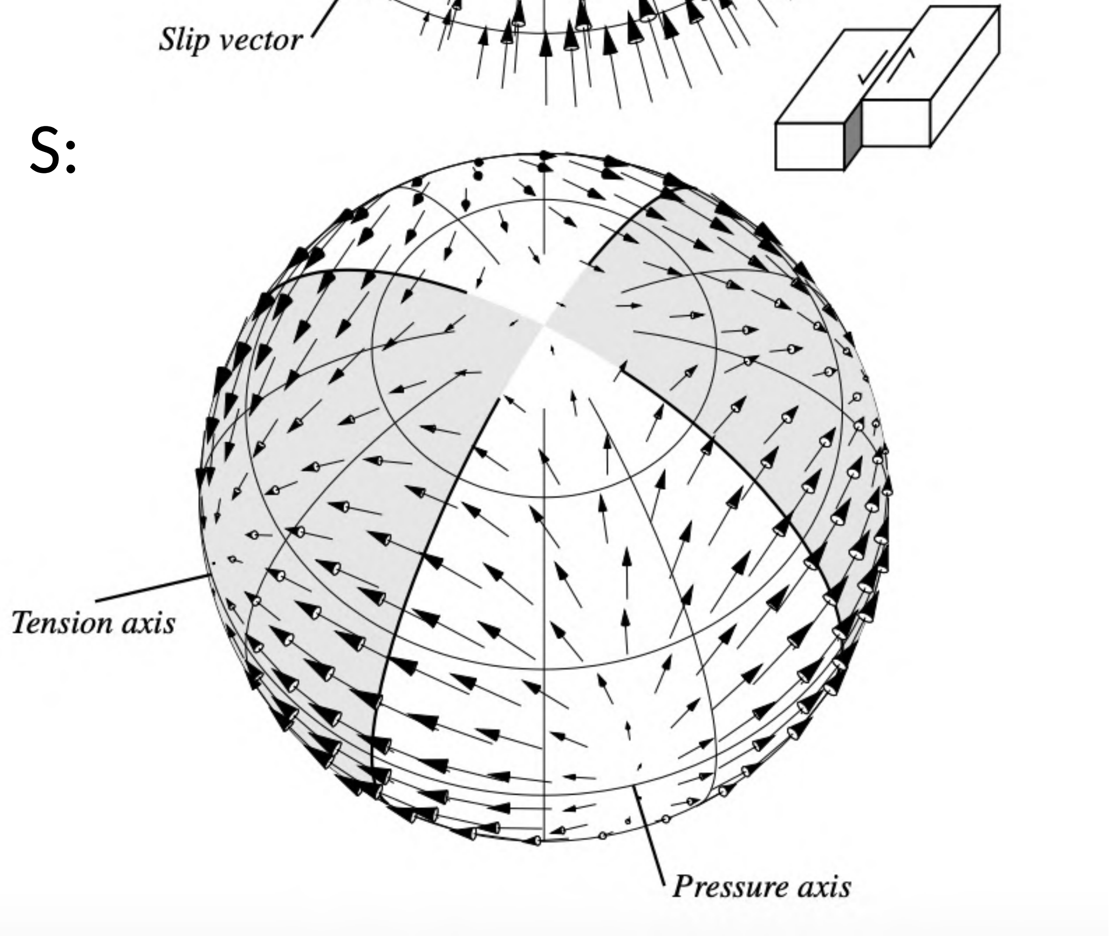
Hudson et al. (2020), JGR

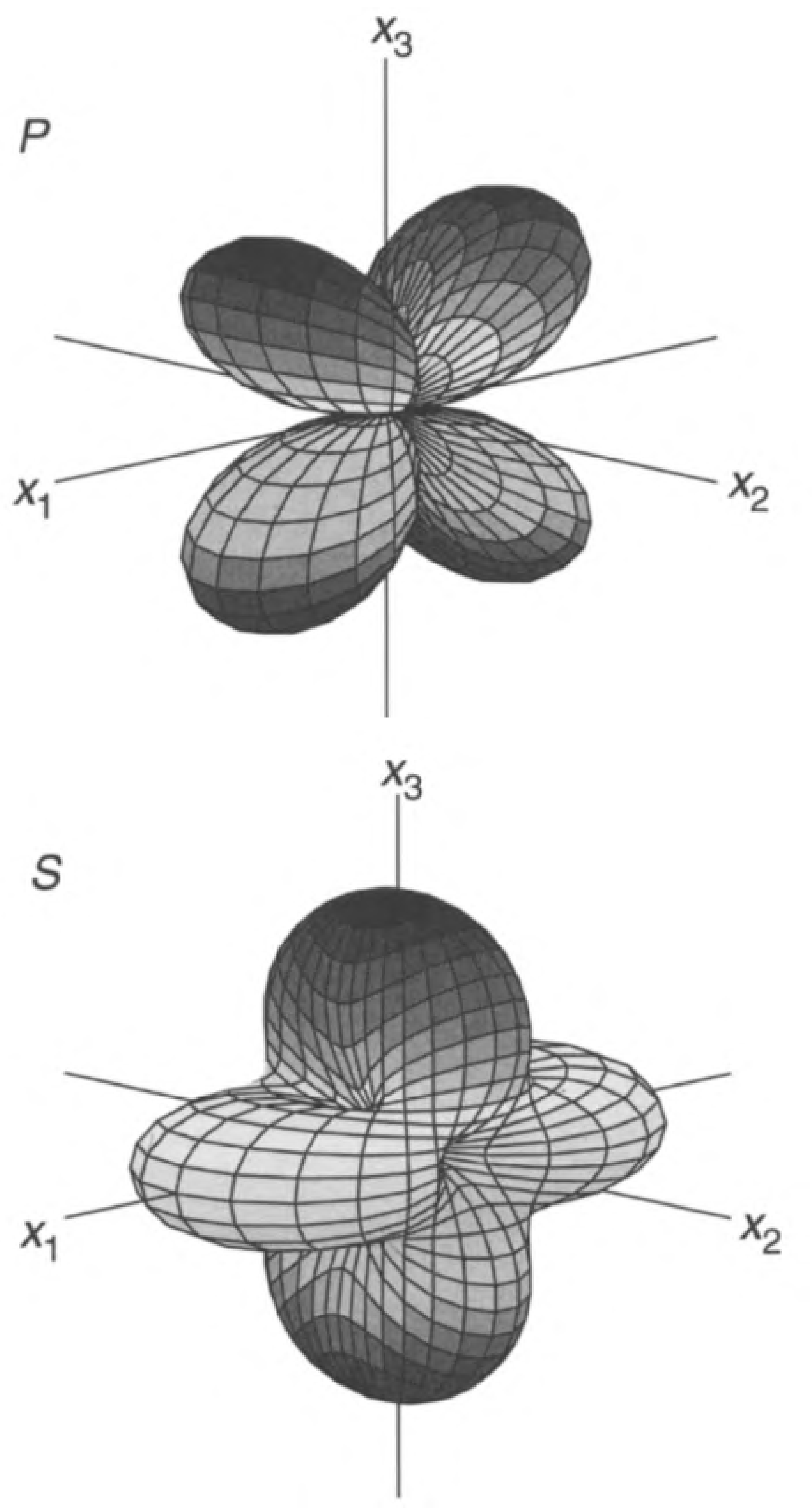
Radiation patterns

P:



S:



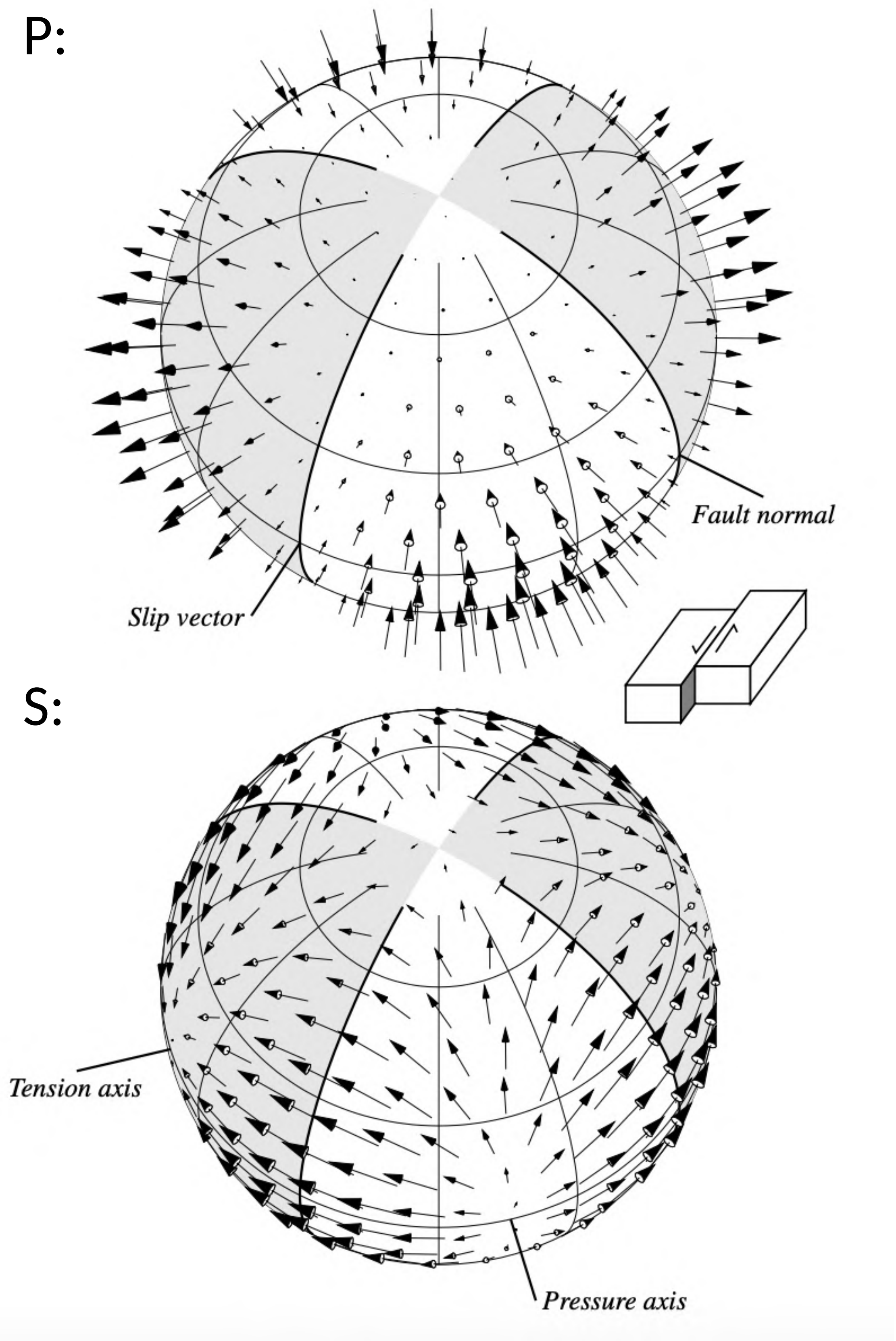


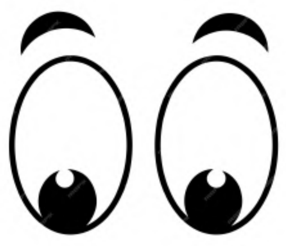
Try Pitch

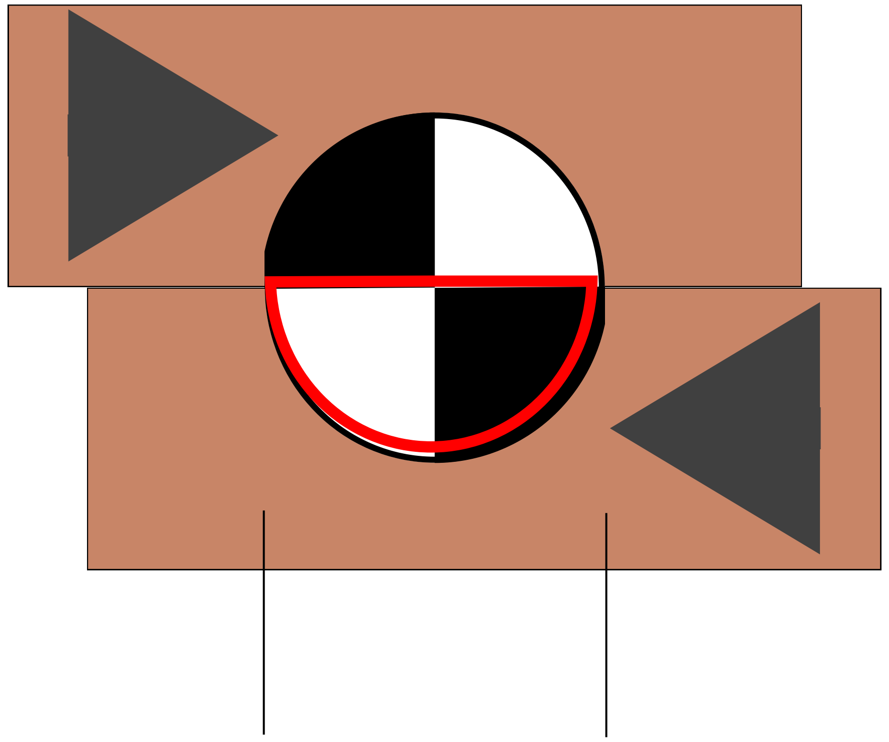
Radiation patterns

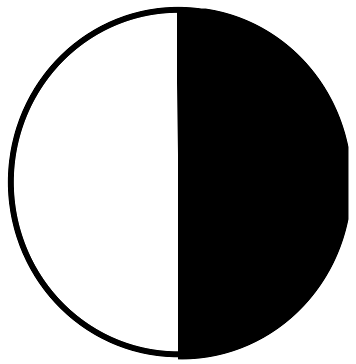
P:

S:





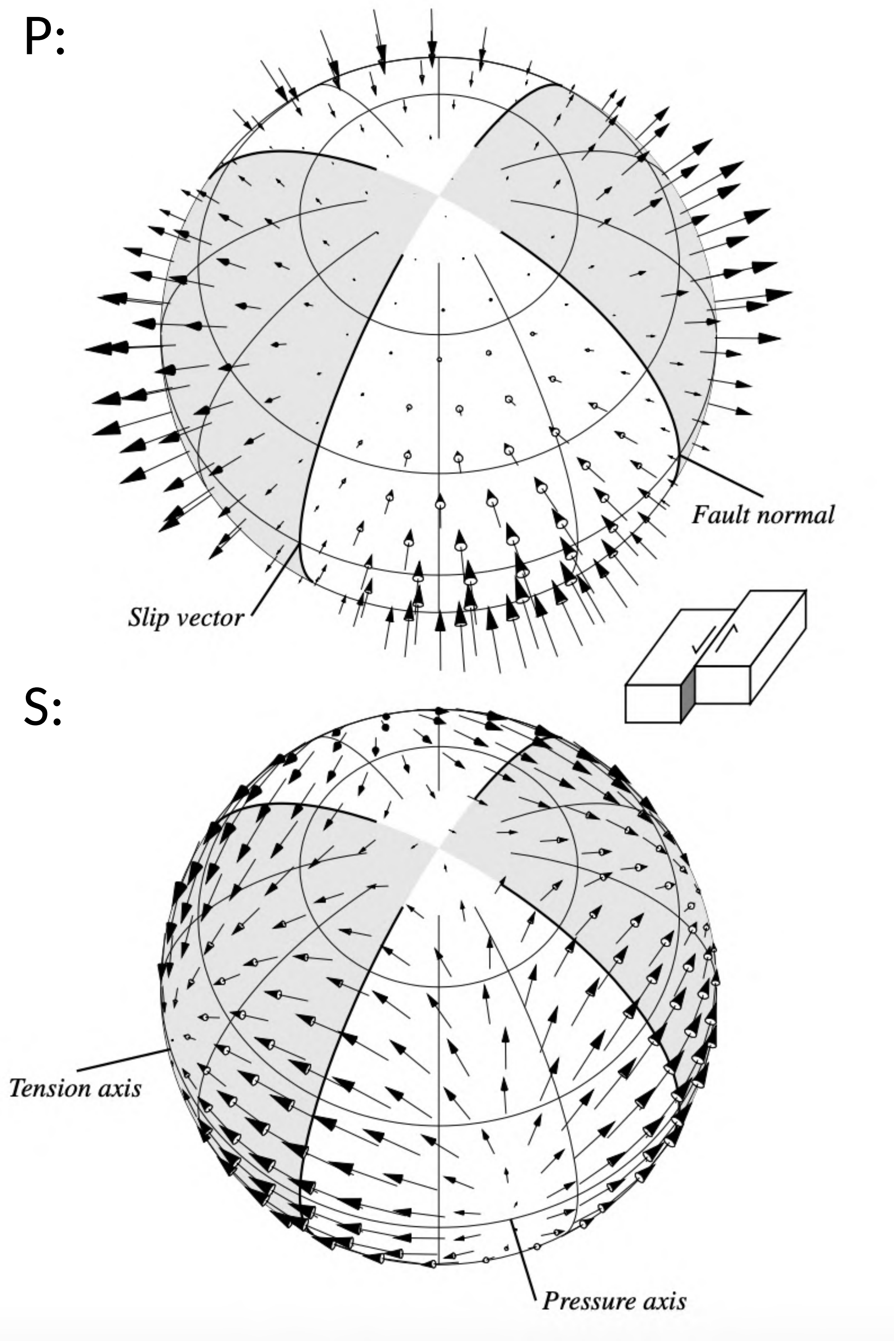


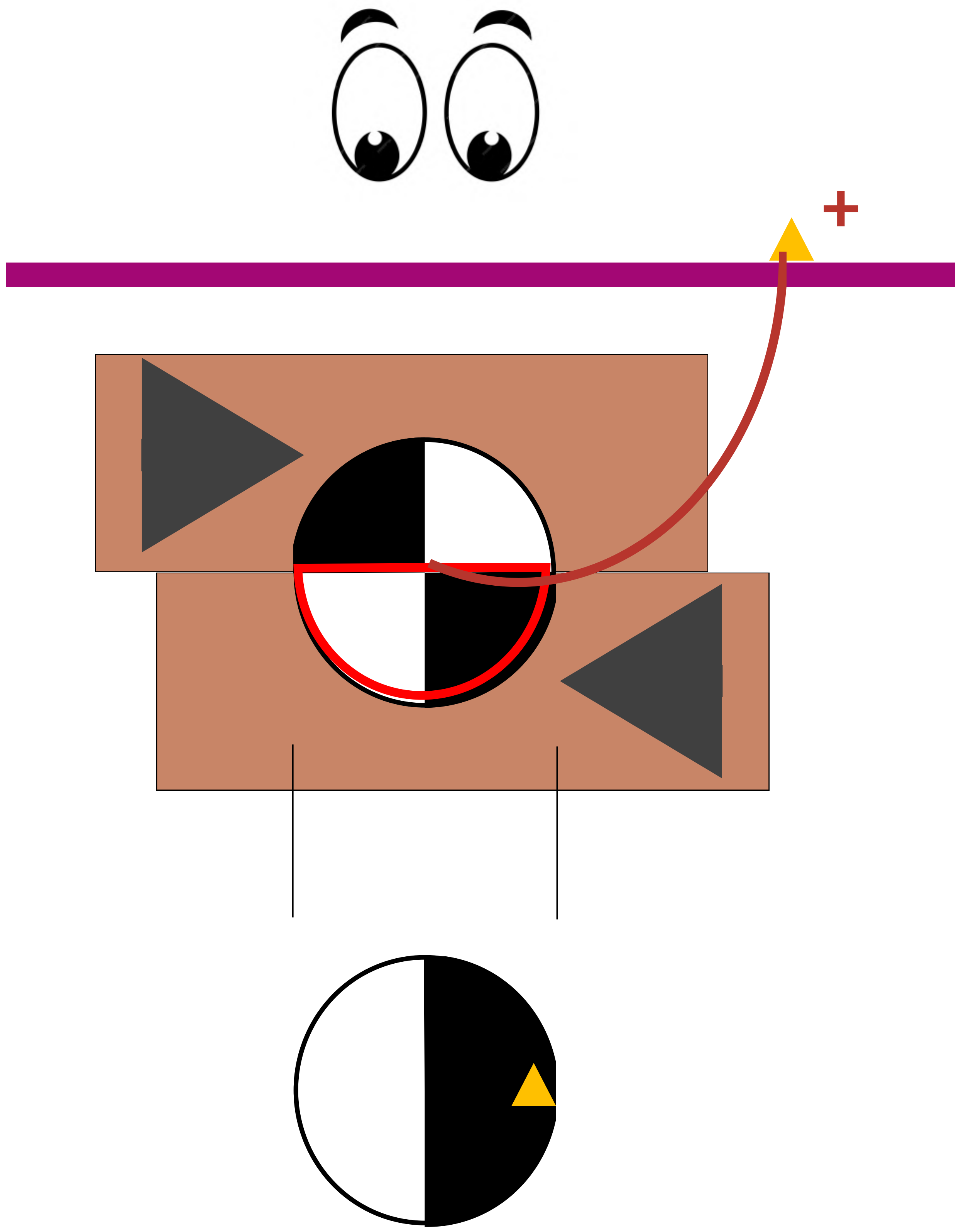


Radiation patterns

P:

S:

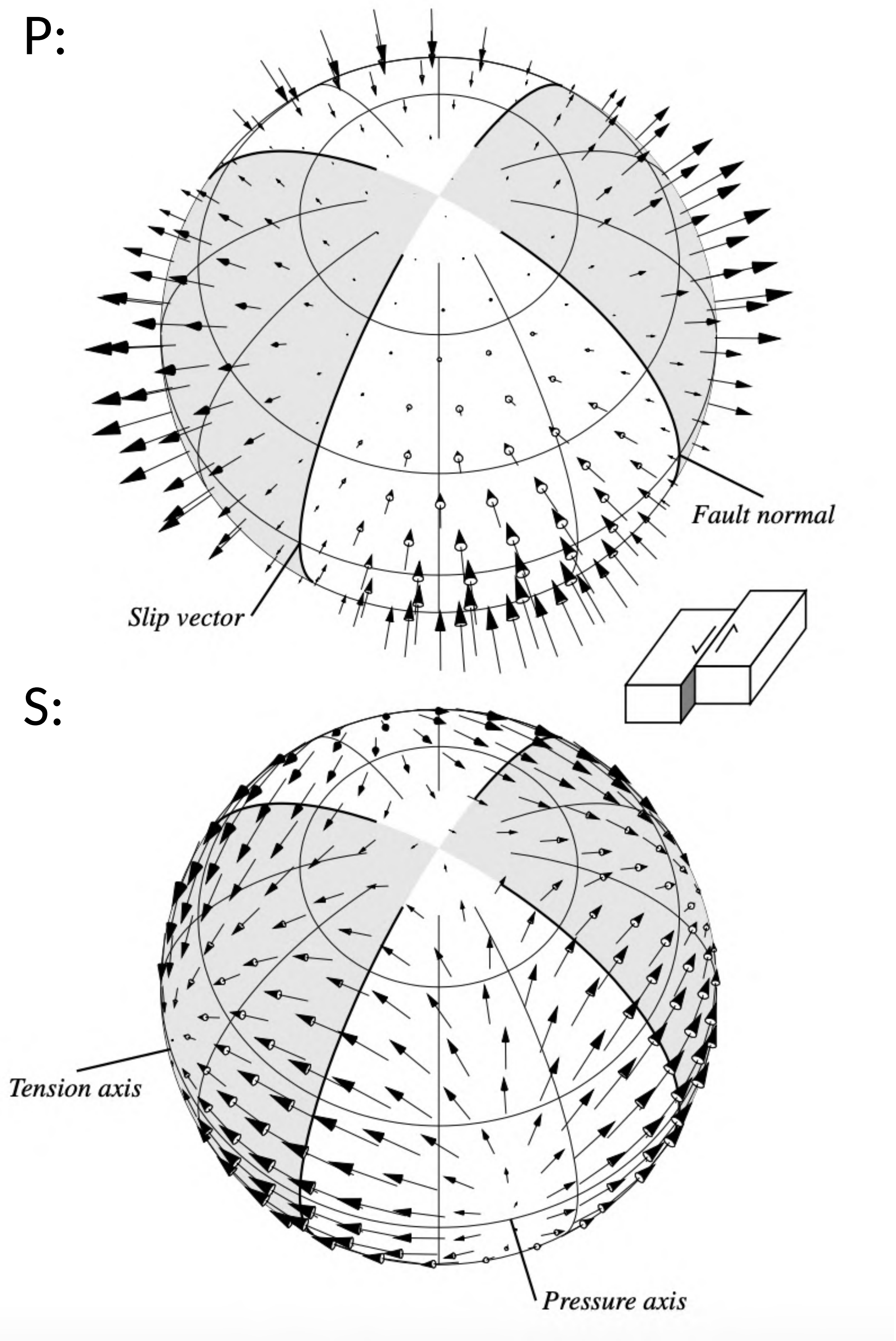


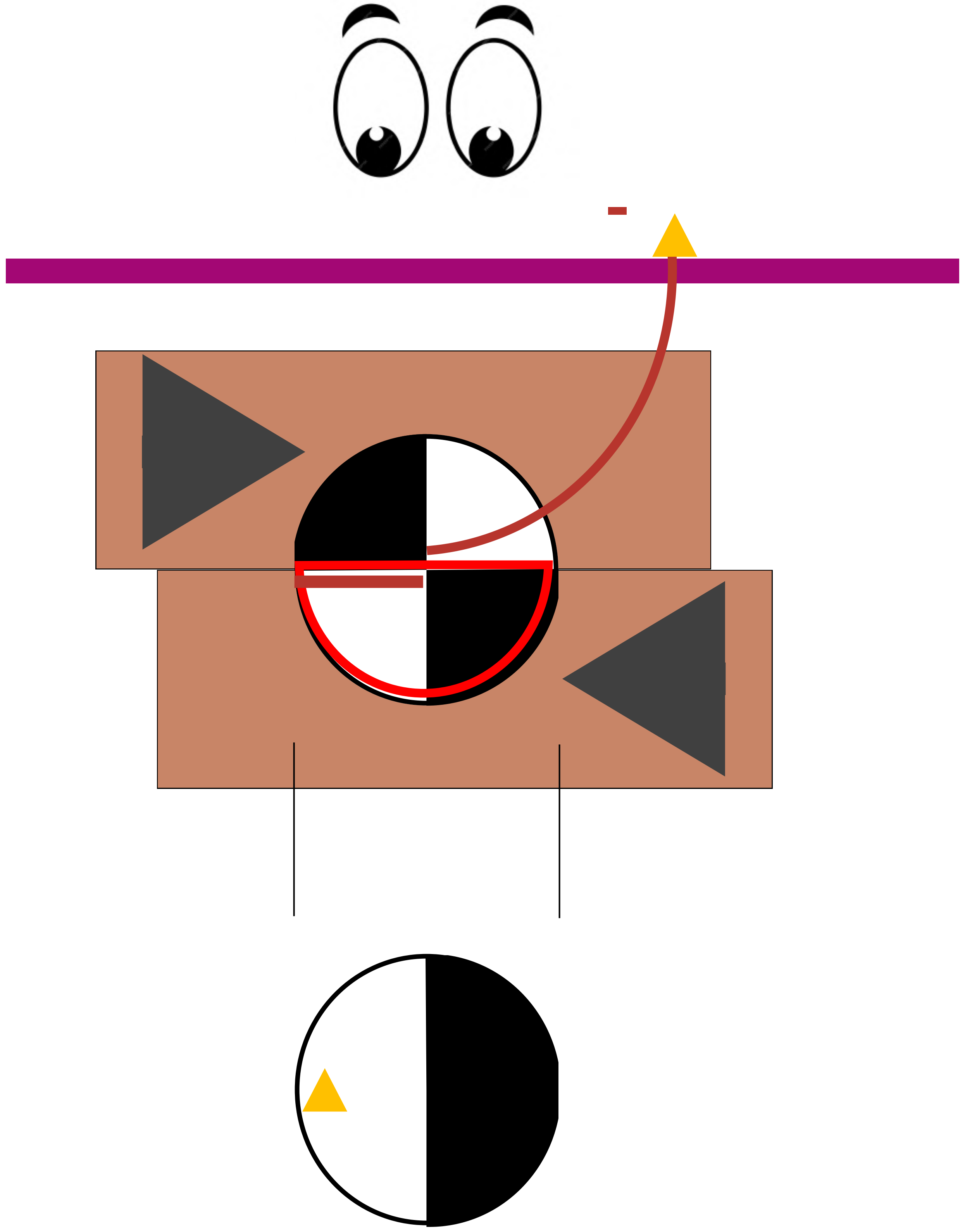


Radiation patterns

P:

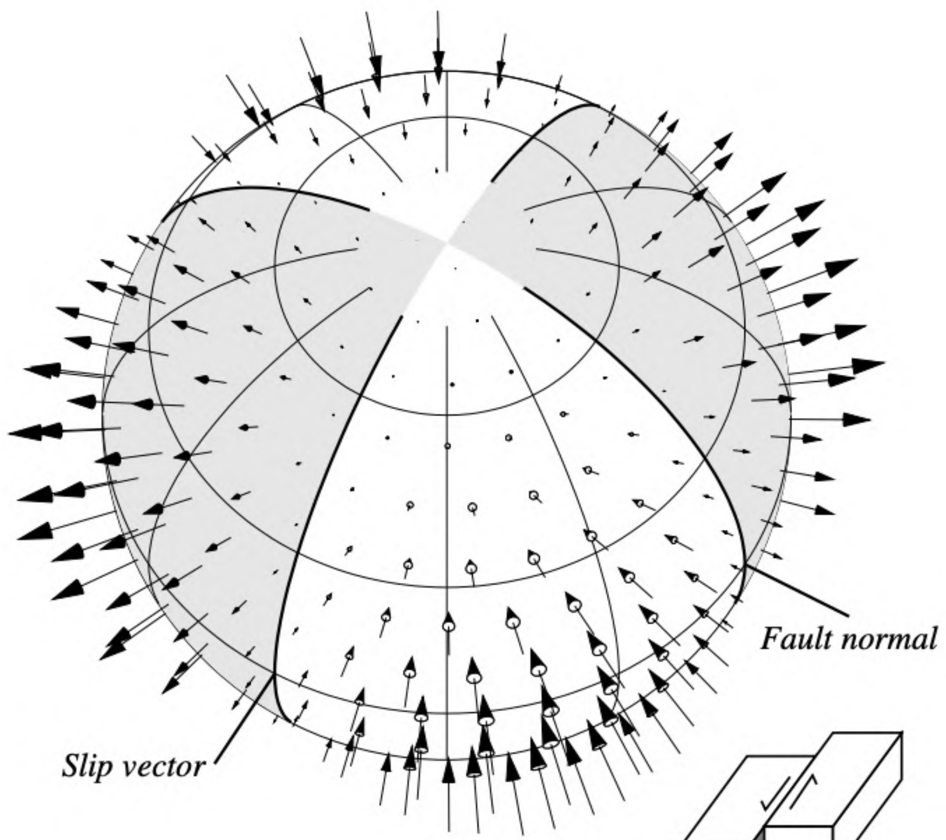
S:



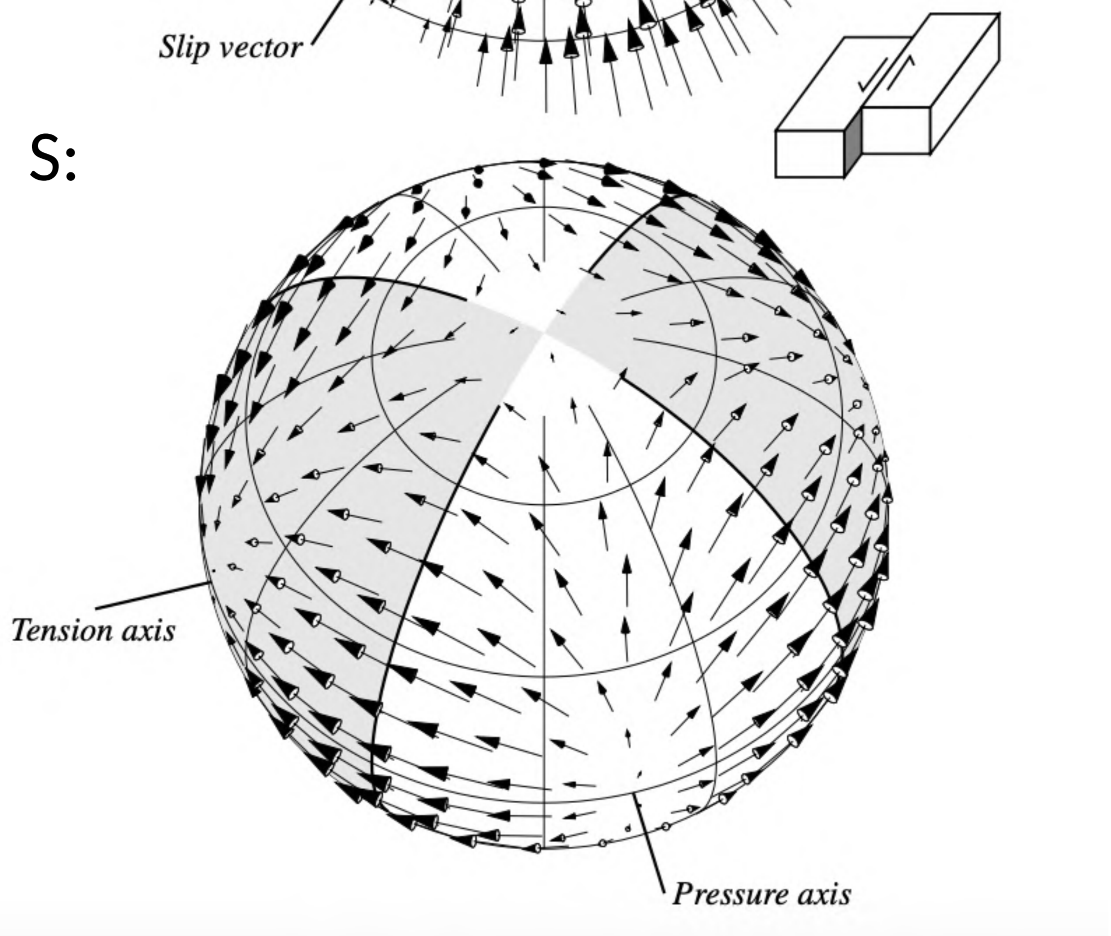


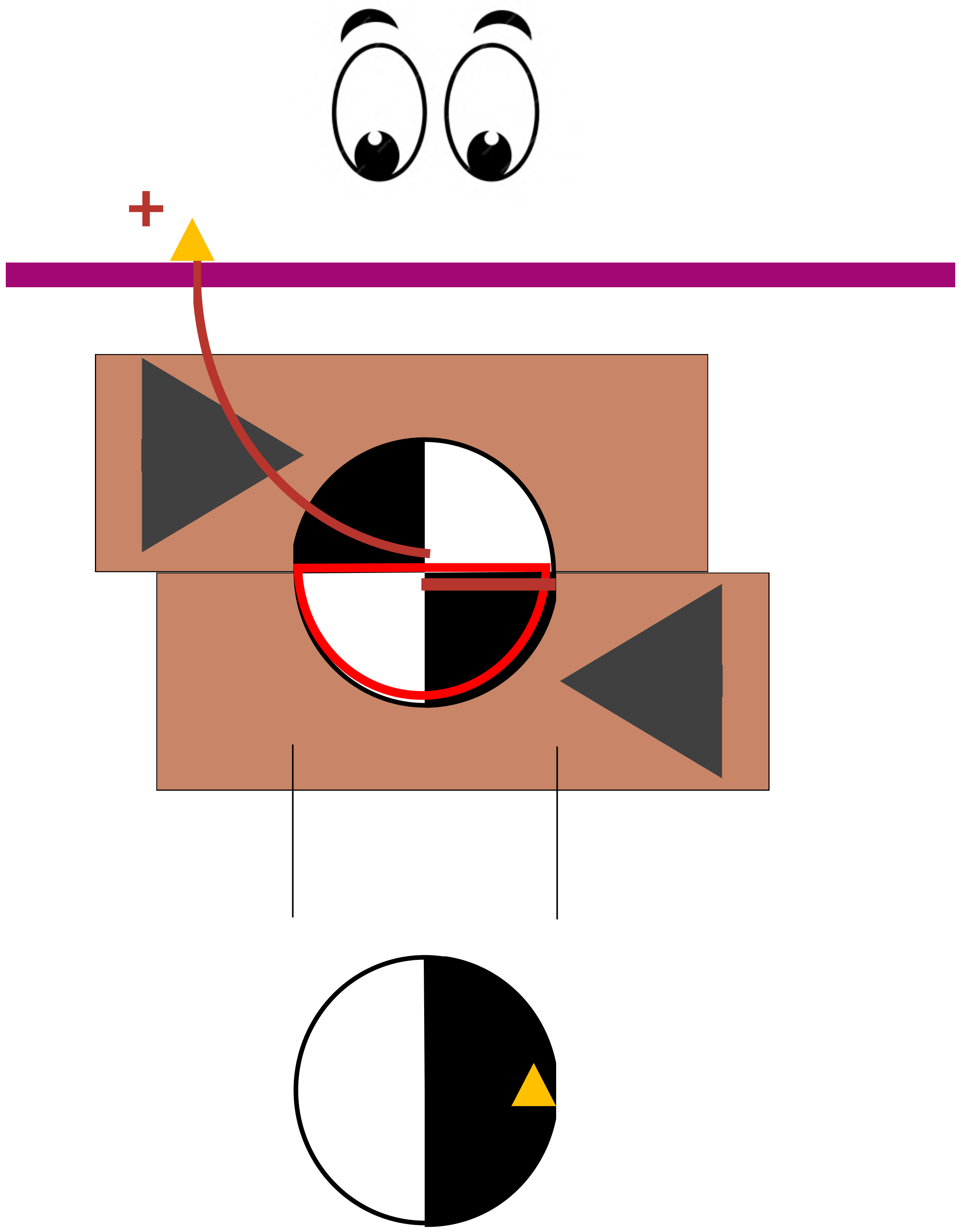
Radiation patterns

P:



S:

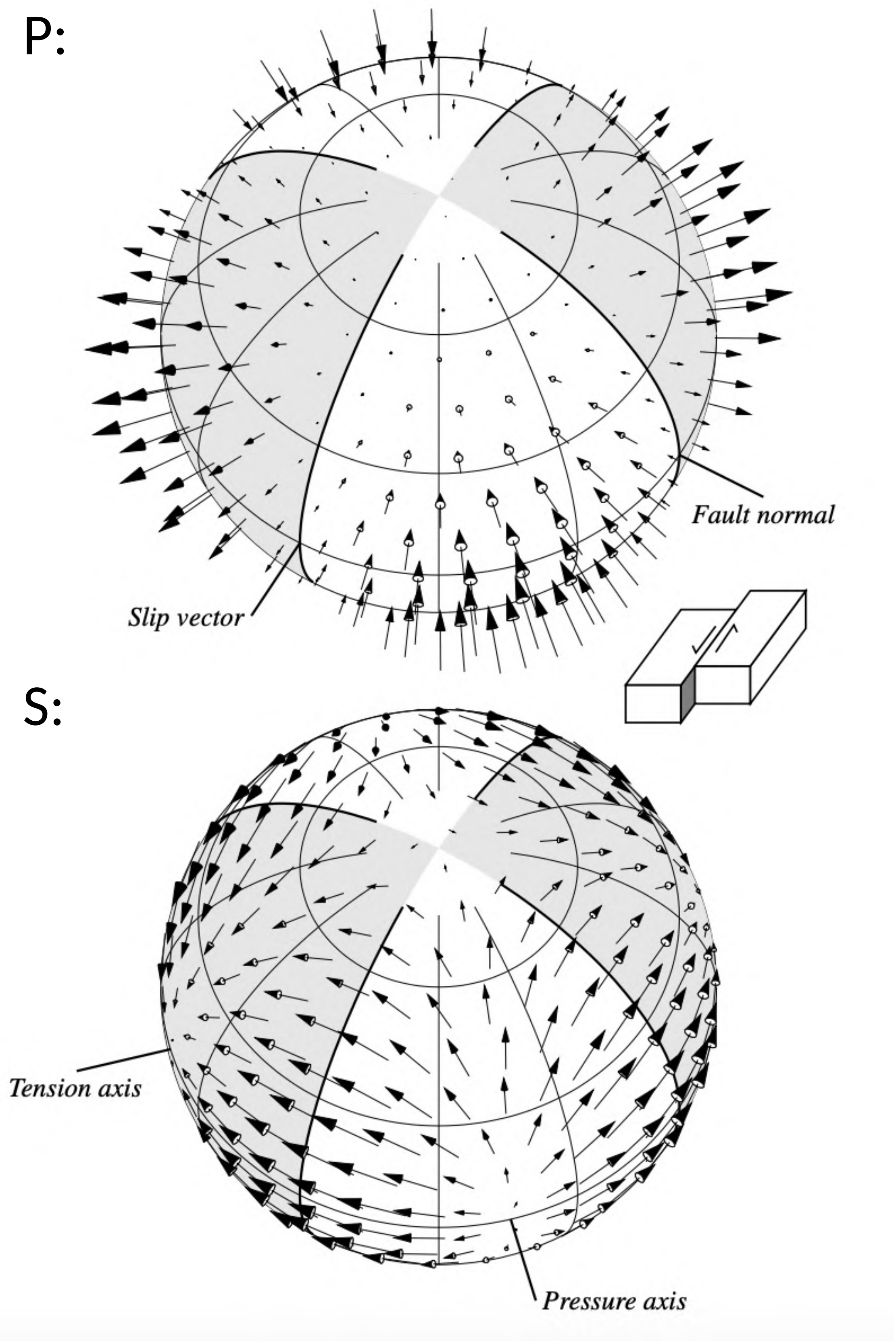


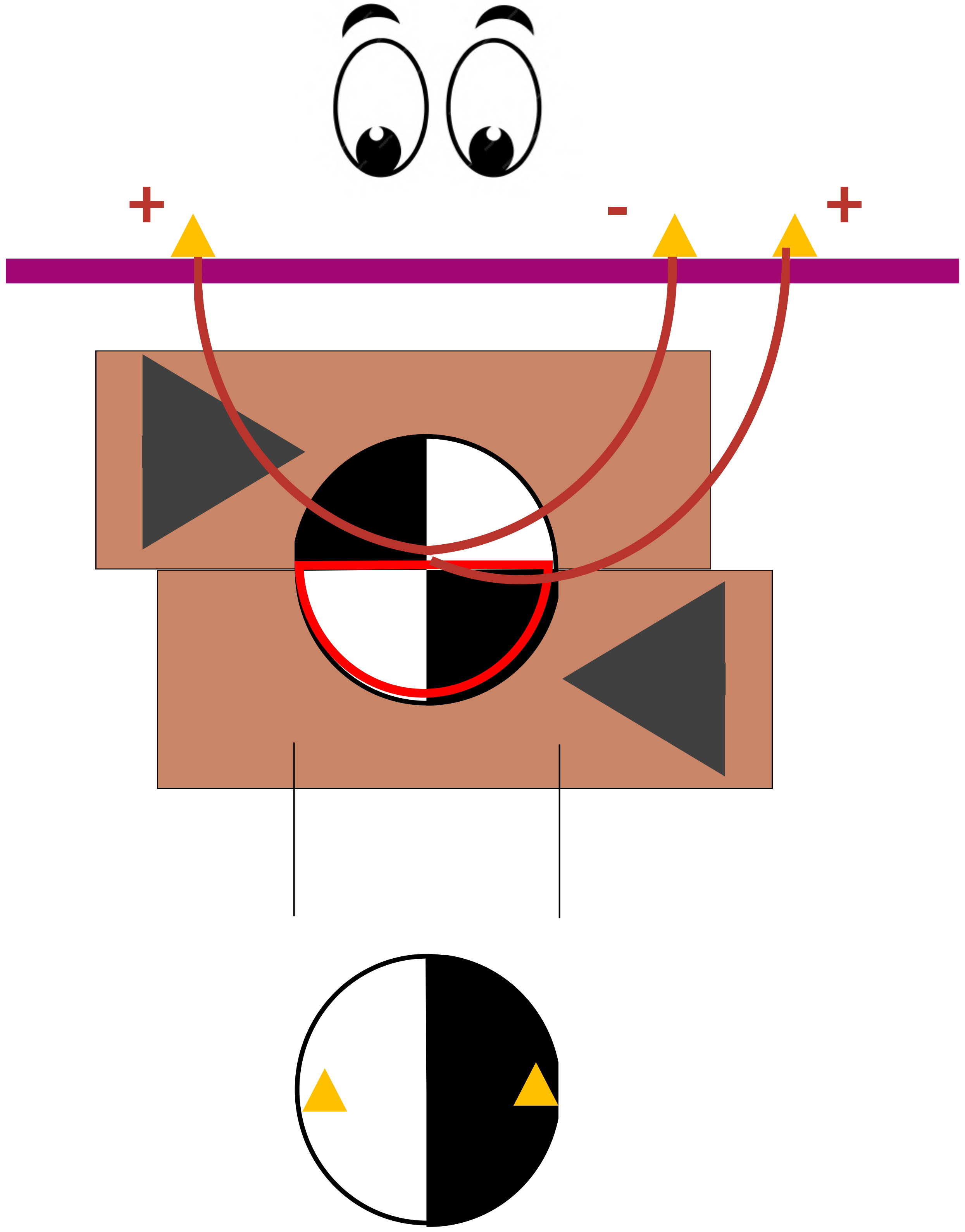


Radiation patterns

P:

S:





Radiation patterns

Lune plot:

Non-double couple sources:

We don't support importing this element yet.

Learn more

We don't support importing this element yet.

Learn more

Try Pitch

ETH zürich

See Tape and Tape (2012) for more details

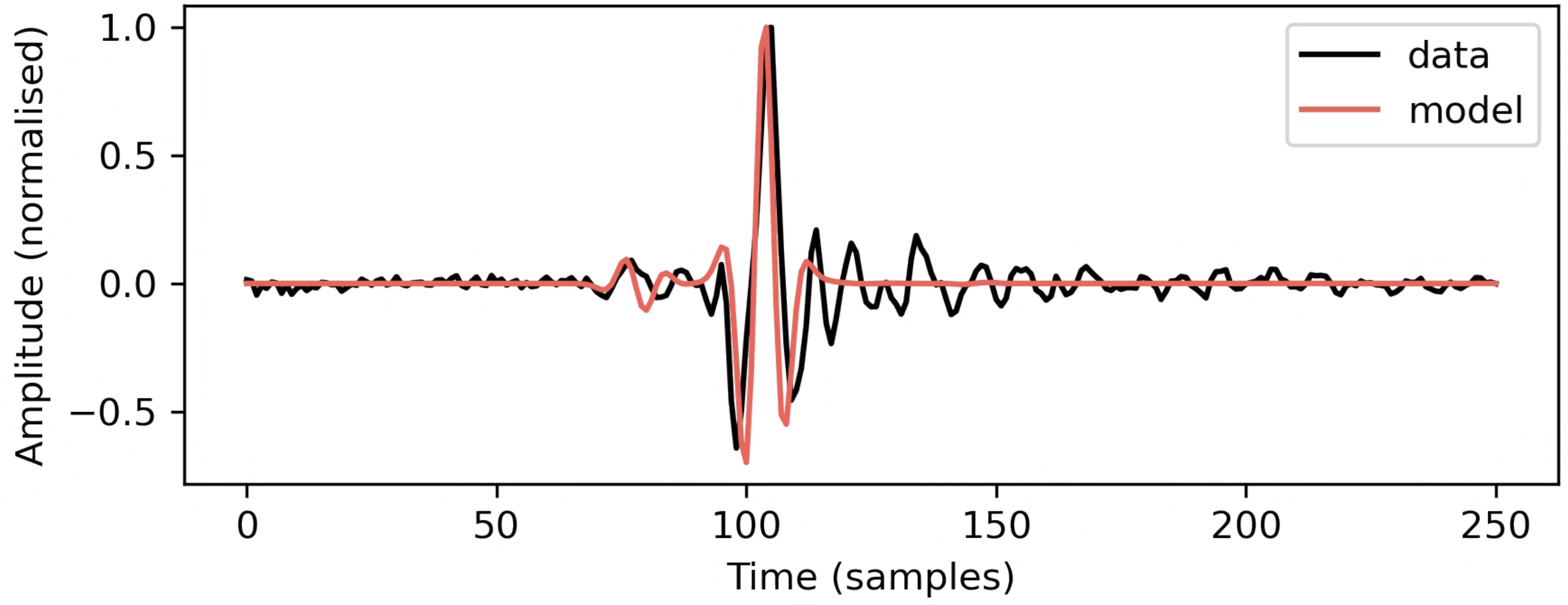
Seismic source mechanism inversion: Introduction

1. Source mechanisms (moment tensors)

2. Data

3. Inversion setup

4. Nuances of fibreoptics



Seismic source mechanism inversion: Introduction

1. Source mechanisms (moment tensors)
2. Data
3. Inversion setup
4. Nuances of fibreoptics

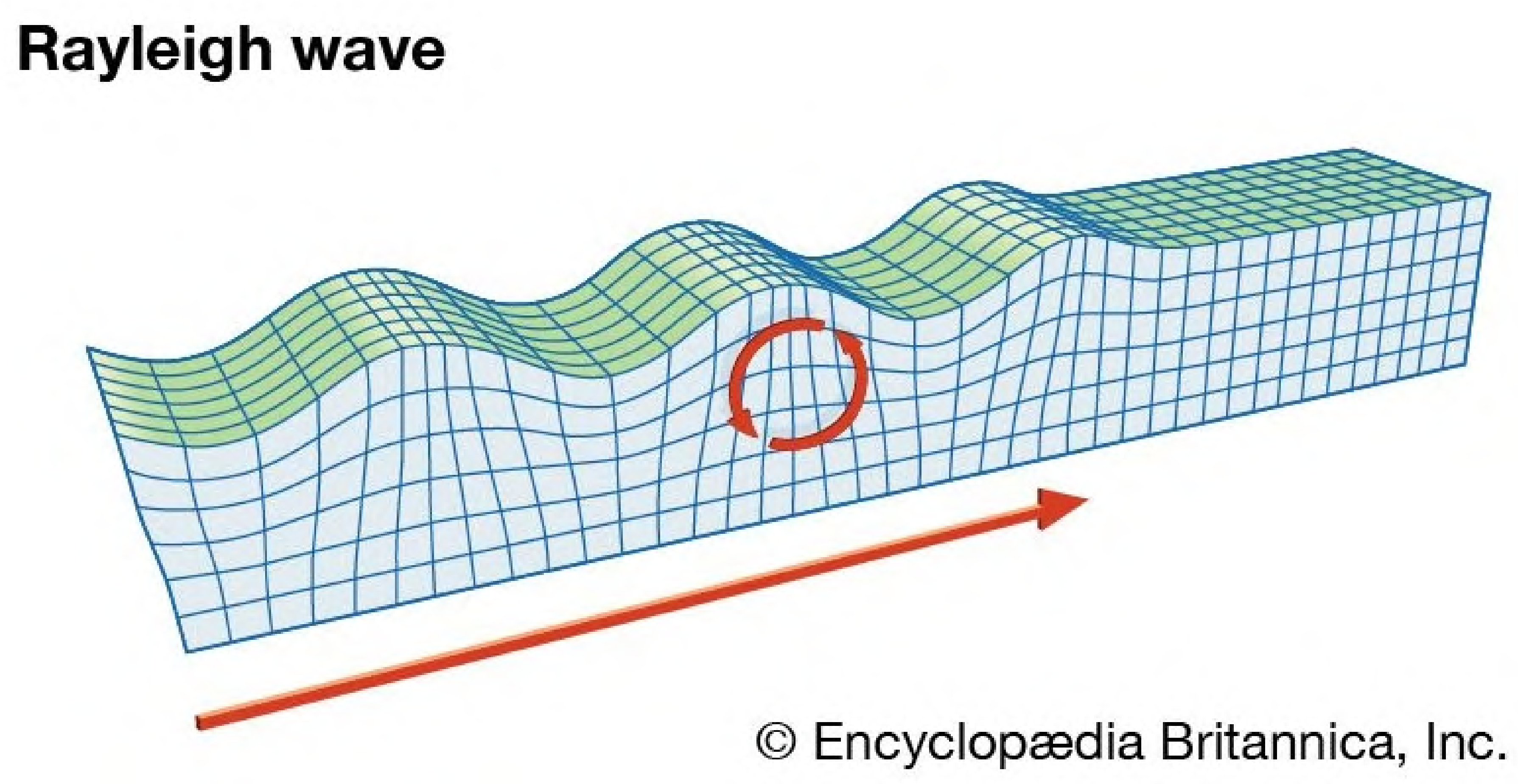
Surface wave velocity anisotropy inversion: A little theory

Rayleigh waves

2. Anisotropy

3. Data

4. Inversion setup



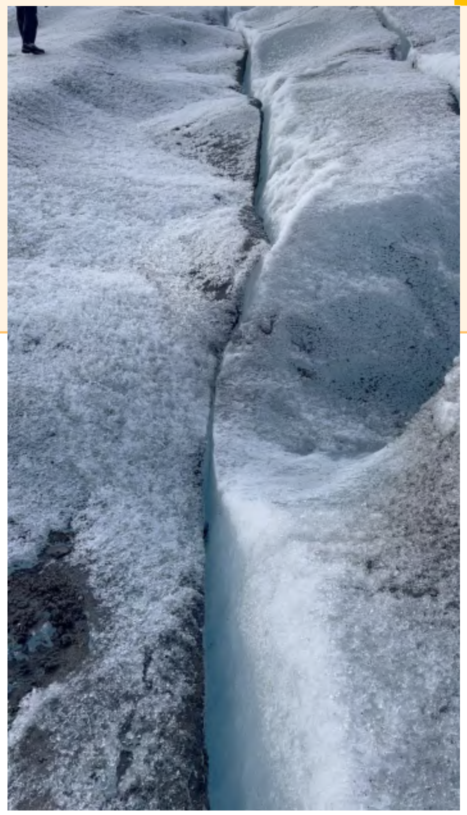
Surface wave velocity anisotropy inversion: A little theory

Rayleigh waves
2. Anisotropy

3. Data

4. Inversion setup

We don't suppo s element yet.

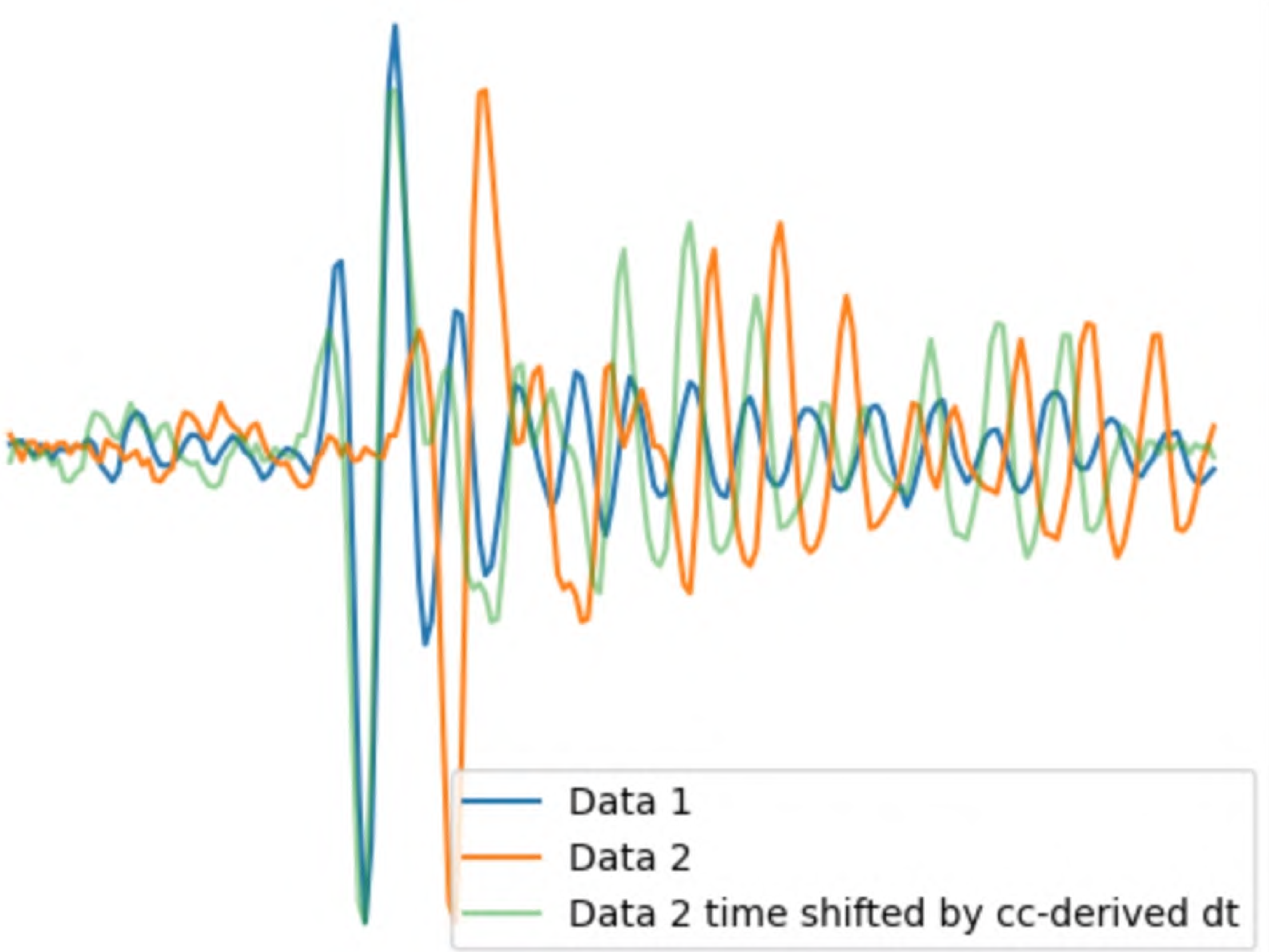


Surface wave velocity anisotropy inversion: A little theory

Rayleigh waves
2. Anisotropy

3. Data

4. Inversion setup



Surface wave velocity anisotropy inversion: A little theory

Rayleigh waves
2. Anisotropy

3. Data

4. Inversion setup

The practical

Seismic source mechanism inversion

Structure:
- Jupyter notebook:
  - Familiarisation with dataset
  - Setup inversion
  - Run inversion
  - * Play around with visualisation of results

Learning aims:
- Exposure to fibreoptic sensing data
- Introduction to linear inversion
- Introduction to full waveform seismic source inversion

Getting started:
- Copy data
- Install any dependencies (NonLinLocPy, SeisSrcInv)
- Open notebook

Surface wave velocity anisotropy inversion

Structure:
- Jupyter notebook:
  - Familiarisation with dataset
  - Setup inversion
  - Run inversion (synthetic + real data)
  - * Further develop yourselves

Learning aims:
- Exposure to fibreoptic sensing data
- Introduction to linear inversion (and a very simple approach to non-linear systems)
- Introduction to glacier anisotropy

Getting started:
- Copy data
- Install any dependencies (NonLinLocPy)
- Open notebook

Time to share ideas...

To conclude…

More familiar with fibreoptic sensing (DAS) data

Have had a fun play with a seismic inversion problem

Want to make a presentation like this one?

Start with a fully customizable template, create a beautiful deck in minutes, then easily share it with anyone.

Create a presentation (It's free)